

JFK University Database Documentation

1. Introduction:

I. Motivation for choosing a particular domain.

We choose a university management system for JAF University because we are university students ourselves. We realize the importance of such a system, in not just the lives of university, but the students as well. For instance, students' information being deleted or corrupted on accident.

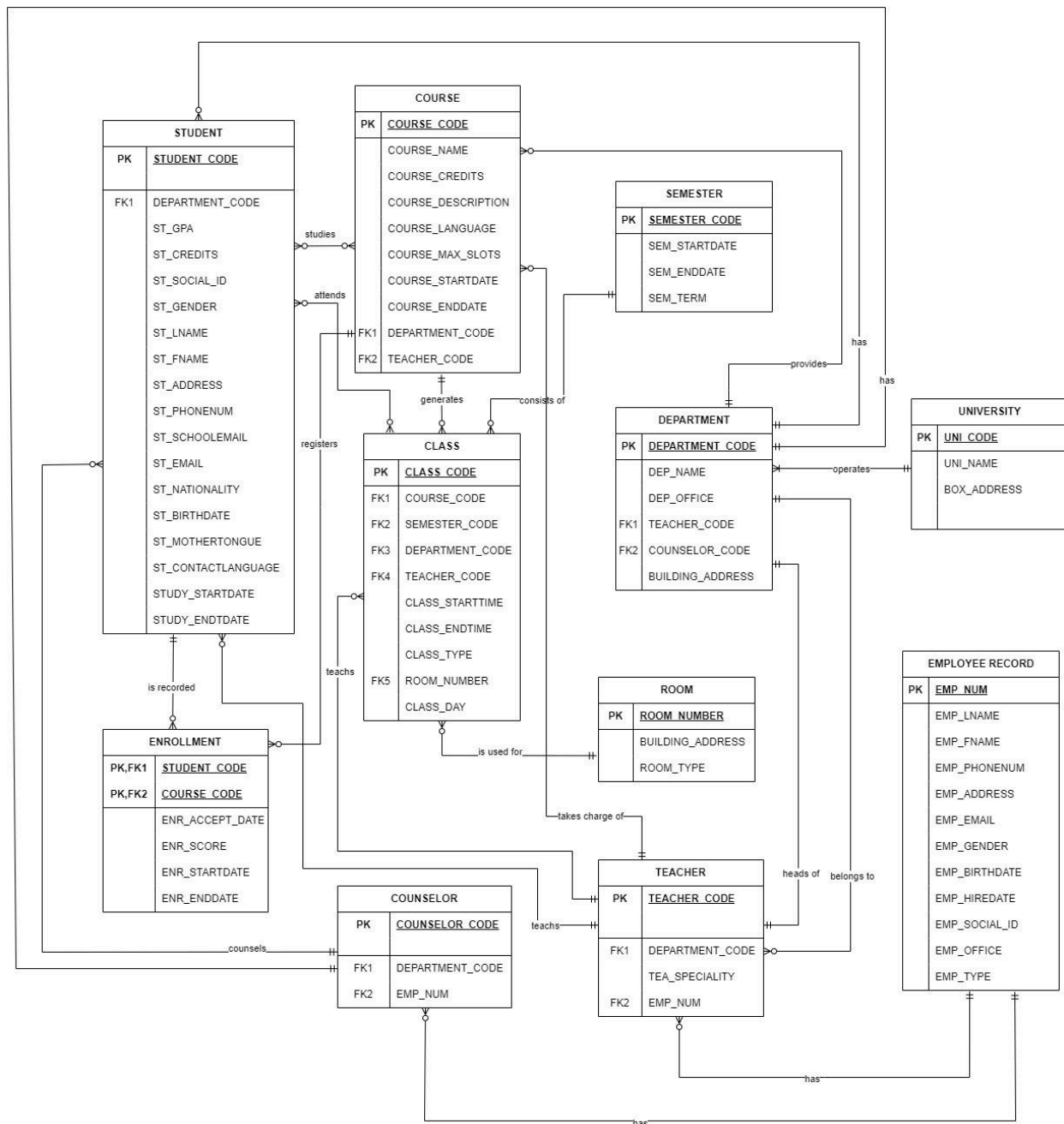
II. The general description of the domain.

The domain we choose is a management system of a University. The data contains students, counselors, teachers, university information, departments, enrollment, courses, classes, semesters, employee record and rooms information. The system allows the university to easily access all this data and collect the needed data at any time.

III. Elicit the business rules of the domain, e.g., involved entities, certain limitations or relationships.

1. The domain is only for students within JAF university, in a specific branch.
2. We will use the english alphabet for our database (e.g students with chinese names will use their pinyin form).
3. Each student belongs to only one department.
4. A student can study many courses and have to make enrollments to access the courses. The enrollment time is available one month until one week before a course starts. A course can be registered by many students. The enrollment is recorded with the course grade at the end of the course, a student may not have course grade if he/she fails the course.
5. A student can attend many classes that belong to the courses he/she enrolls in.
6. A course can be generated into many classes with different types/purposes such as teaching or tutoring. Specific class belongs to only one course.
7. Class schedules are arranged within the semester in different terms. One class is arranged to be taught in one specific room. A room can be occupied by many classes at different times.
8. Each department has only one counselor to counsel the students belonging to the department, one head teacher that leads the department. Each department has many students and many teachers. One counselor belongs to only one department. A department can provide many courses, and every course belongs to only one department.
9. A teacher can take charge of several courses, and a course is taught by one teacher. The teacher goes to the classes to teach students the course. Each teacher belongs to only one department.
10. JFK university has many departments, and all those departments only belong to JFK university.
11. Every employee in the university has an employee record. The employee record system includes all records of the university employee.

2. The ERD of the real-world problem.



3. The relational model, transformed from the ERD.

STUDENT(STUDENT_CODE, DEPARTURE_CODE, ST_GPA, ST_CREDITS, ST_SOCIAL_ID, ST_GENDER, ST_LNAME, ST_FNAME, ST_ADDRESS, ST_PHONENUM, ST_SCHOOLEMAIL, ST_EMAIL, ST_NATIONALITY, ST_BIRTHDATE, ST_MOTHERTONGUE, ST_CONTACTLANGUAGE, STUDY_STARTDATE, STUDY_ENDDATE)

CLASS(CLASS_CODE, COURSE_CODE, SEMESTER_CODE, DEPARTMENT_CODE, TEACHER_CODE, CLASS_STARTTIME, CLASS_ENDTIME, CLASS_TYPE, ROOM_NUMBER, CLASS_DAY)

DEPARTMENT(DEPARTMENT_CODE, DEP_NAME, DEP_OFFICE, TEACHER_CODE, COUNSELOR_CODE, BUILDING_ADDRESS)

UNIVERSITY(UNI_CODE, UNI_NAME, BOX_ADDRESS)

COURSE(COURSE_CODE, COURSE_NAME, COURSE_CREDITS, COURSE_DESCRIPTION, COURSE_LANGUAGE, COURSE_MAX_SLOTS, COURSE_STARTDATE, COURSE_ENDDATE, DEPARTMENT_CODE, TEACHER_CODE)

SEMESTER(SEMESTER_CODE, SEM_STARTDATE, SEM_ENDDATE, SEM_TERM)

ENROLMENT(STUDENT_NUMBER, COURSE_CODE, ENR_ACCEPT_DATE, ENR_SCORE, ENR_STARTDATE, ENR_ENDDATE)

COUNSELOR(COUNSELOR_CODE, DEPARTMENT_CODE, EMP_NUM)

ROOM(ROOM_NUMBER, BUILDING_ADDRESS, ROOM_TYPE)

TEACHER(TEACHER_CODE, DEPARTMENT_CODE, TEA_SPECIALITY, EMP_NUM)

EMPLOYEE RECORD(EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_PHONENUM, EMP_ADDRESS, EMP_EMAIL, EMP_GENDER, EMP_BIRTHDATE, EMP_HIREDATE, EMP_SOCIAL_ID, EMP_OFFICE, EMP_TYPE)

4. Certain improvements

We have applied normalization when developing ERD to improve the database structure, below are some examples:

1NF(COUNSELOR_CODE, DEPARTMENT_CODE, EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_PHONENUM, EMP_ADDRESS, EMP_EMAIL, EMP_GENDER, EMP_BIRTHDATE, EMP_HIREDATE, EMP_SOCIAL_ID, EMP_OFFICE, EMP_TYPE)

1NF(TEACHER_CODE, DEPARTMENT_CODE, TEA_SPECIALITY, EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_PHONENUM, EMP_ADDRESS, EMP_EMAIL, EMP_GENDER, EMP_BIRTHDATE, EMP_HIREDATE, EMP_SOCIAL_ID, EMP_OFFICE, EMP_TYPE)

1NF(CLASS_CODE, COURSE_CODE, SEMESTER_CODE, DEPARTMENT_CODE, TEACHER_CODE, CLASS_STARTTIME, CLASS_ENDTIME, CLASS_TYPE, ROOM_NUMBER, BUILDING_ADDRESS, ROOM_TYPE, CLASS_DAY)

From First normal form 1NF, eliminating transitive dependencies to achieve Third Normal Form 3NF:

3NF(COUNSELOR_CODE, DEPARTMENT_CODE, EMP_NUM)

3NF(TEACHER_CODE, DEPARTMENT_CODE, TEA_SPECIALITY, EMP_NUM)

3NF(EMP_NUM, EMP_LNAME, EMP_FNAME, EMP_PHONENUM, EMP_ADDRESS, EMP_EMAIL, EMP_GENDER, EMP_BIRTHDATE, EMP_HIREDATE, EMP_SOCIAL_ID, EMP_OFFICE, EMP_TYPE)

3NF(CLASS_CODE, COURSE_CODE, SEMESTER_CODE, DEPARTMENT_CODE, TEACHER_CODE, CLASS_STARTTIME, CLASS_ENDTIME, CLASS_TYPE, ROOM_NUMBER, CLASS_DAY)

3NF(ROOM_NUMBER, BUILDING_ADDRESS, ROOM_TYPE)

Or from First normal form 1NF of full Enrollment table including students and course details to transfer to Second Normal Form 2NF enrollment table with two primary keys, student and course have separated tables as shown in the relation model above:

2NF(STUDENT_NUMBER, COURSE_CODE, ENR_ACCEPT_DATE, ENR_SCORE, ENR_STARTDATE, ENR_ENDDATE)

5. Description of the application interacting with the database.

I. We have chosen to code Java programming language.

II. A list of the SQL queries implemented in the program.

"select *from emplyee_record"

"select *from class where class_type = 'TEACHING' and day = 'MONDAY'"

"select *from class"

"select *from course where course_language = 'English'"

"select *from course"

"select *from student"

"select *from student where study_startdate = '2022--08-01'"

"select *from student where st_gpa > 3.50"

"select *from semester"

"select *from university"

"select *from enrollment"

"select *from counselor"

"select *from teacher"

"select *from room"

"select *from departement"

III. The first end-user requirement that we are meeting is the ease of use. The user doesn't need technical knowledge in order to retrieve or to add data from the database. Our program is flexible and scalable, as the database supports the use of different data types and accommodates data accumulation.